QuAIR Research Notes **2026**

Note 3: Computation of DLA and noise theory

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3.1 Summary of Progress

This weekly note

1. Establishes a complete hierarchy for DLA computation, including dimension, basis, structure, and $\mathfrak{g}$-purity.
2. Proposes and implements numerical and symbolic algorithms for DLA closure, with complexity analysis.
3. Computes DLA dimensions for three key spin models and identifies open problems for general n.
4. Reviews quantum operations, noise channels, and fundamental operator identities for future control analysis.

3.2 Computation of DLA

3.2.1 Classification of computation problem for DLA

There are distinct levels of computing the DLA as follows.

Definition 1 (DLA Dimension Computation) Given $\mathcal{G} = \{H_1, \ldots, H_k\} \subset \mathrm{Herm}(N)$ and its DLA $\mathfrak{g} = \langle i\mathcal{G}\rangle_{\mathrm{Lie}}$, the **DLA dimension computation** asks only for the real dimension

$$\dim_{\mathbb{R}} \mathfrak{g}.$$

Definition 2 (DLA Basis / Closure Computation) Given $\mathcal{G} = \{H_1, \ldots, H_k\} \subset \mathrm{Herm}(N)$ and its DLA $\mathfrak{g} = \langle i\mathcal{G}\rangle_{\mathrm{Lie}}$, the *DLA basis computation* (closure computation) finds a real basis $\{Y_1, \ldots, Y_d\} \subset \mathfrak{u}(N)$ such that

$$\mathfrak{g} = \mathrm{span}_{\mathbb{R}}\{Y_1, \ldots, Y_d\}.$$

This is the standard form of computing the DLA in the literature.

Definition 3 $\mathfrak{g}$-Purity (Generalized Purity)

- Let $\mathfrak{g} \subseteq \mathfrak{u}(2^n)$ be a subalgebra, with complexification $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$.
- Let $\{B_j\}_{j=1}^{\dim(\mathfrak{g})}$ be an orthonormal basis of $\mathfrak{g}_{\mathbb{C}}$ under the inner product $\langle A, B\rangle = \mathrm{Tr}[A^\dagger B]$.

For any Hermitian operator $H \in i\mathfrak{u}(2^n)$, the **$\mathfrak{g}$-purity** of H is defined as

$$\mathcal{P}_{\mathfrak{g}}(H) = \mathrm{Tr}\left[H_{\mathfrak{g}}^2\right] = \sum_{j=1}^{\dim(\mathfrak{g})} \left|\mathrm{Tr}\left[B_j^\dagger H\right]\right|^2,$$

where $H_{\mathfrak{g}}$ denotes the orthogonal projection of H onto $\mathfrak{g}_{\mathbb{C}}$.

Definition 4 (DLA Structure Identification) Given $\mathcal{G} = \{H_1, \ldots, H_k\} \subset \mathrm{Herm}(N)$ and its DLA $\mathfrak{g} = \langle i\mathcal{G}\rangle_{\mathrm{Lie}}$, **DLA structure identification** determines the Lie-algebra isomorphism class of $\mathfrak{g}$, including its decomposition into center and simple summands:$\mathfrak{g} \cong \mathfrak{z}(\mathfrak{g}) \oplus \mathfrak{s}_1 \oplus \cdots \oplus \mathfrak{s}_m$.

Let $\mathfrak{g} = \mathfrak{g}_{\mathrm{DLA}}$ be the dynamical Lie algebra with decomposition

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_{k-1} \oplus \mathfrak{g}_k,$$

The variance of $\ell_\theta(\rho, O)$ is

$$\mathrm{Var}_\theta\left[\ell_\theta(\rho, O)\right] = \sum_{j=1}^{k-1} \frac{\mathcal{P}_{\mathfrak{g}_j}(\rho)\mathcal{P}_{\mathfrak{g}_j}(O)}{\dim(\mathfrak{g}_j)}.$$

We further distinguish two computational paradigms:

- *Symbolic DLA computation*: Uses exact commutation relations and symbolic matrix expressions, yielding error-free results.
- *Numerical DLA computation*: Uses floating-point matrix representations and numerical rank detection, trading precision for speed.

Theorem 5 ((DLA Construction via Inductive Closure):) *Let $\{X_1, \ldots, X_k\} \subset \mathfrak{u}(N)$ be a finite set of skew-Hermitian generators. Define an inductive sequence of real subspaces of $\mathfrak{u}(N)$:*

$$\begin{cases} \mathfrak{g}_0 = \mathrm{span}_{\mathbb{R}}\{X_1, X_2, \ldots, X_k\} \\ \mathfrak{g}_{m+1} = \mathfrak{g}_m + \mathrm{span}_{\mathbb{R}}\big\{[A, B] \mid A, B \in \mathfrak{g}_m\big\} \quad (\forall m \geq 0) \end{cases}$$

where $\mathrm{span}_{\mathbb{R}}\, S$ denotes the real linear span of a set S. Then:

1. *The sequence stabilizes: There exists a minimal integer $M \geq 0$ such that $\mathfrak{g}_{M+1} = \mathfrak{g}_M$;*
2. *The stable subspace $\mathfrak{g}_M$ is exactly the DLA $\mathfrak{g}_{\mathrm{DLA}}$.*

This algorithm computes the dynamical Lie algebra (DLA) by iteratively closing a generating set under the Lie bracket. It only maintains a real linearly independent basis, avoiding explicit representation of infinite subspaces $\mathfrak{g}_m$.

Algorithm 1 Computable Inductive Closure Algorithm for DLA

Require: Skew-Hermitian generators $\{X_1, X_2, \ldots, X_k\} \subset \mathfrak{u}(N)$
Ensure: Real linearly independent basis $\mathcal{B}$ of $\mathfrak{g}_{\mathrm{DLA}}$
$\mathcal{B} \leftarrow$ real-linearly independent subset of $\{X_1, \ldots, X_k\}$
while True **do**
  $\mathcal{C} \leftarrow \big\{AB - BA \;\big|\; A, B \in \mathcal{B}\big\}$
  $\mathcal{T} \leftarrow \mathcal{B} \cup \mathcal{C}$
  $\mathcal{B}_{\mathrm{new}} \leftarrow$ real-linearly independent subset of $\mathcal{T}$
  if $|\mathcal{B}_{\mathrm{new}}| = |\mathcal{B}|$ **then**
    $\mathcal{B} \leftarrow \mathcal{B}_{\mathrm{new}}$
    break
  end if
  $\mathcal{B} \leftarrow \mathcal{B}_{\mathrm{new}}$
end while
return $\mathcal{B}$

The algorithm terminates in finite steps since $\mathfrak{u}(N)$ is finite-dimensional. Each iteration computes $O(d^2)$ Lie brackets and performs real linear algebra reduction.

Time complexity: $O(Md^2N^3 + Md^3N^2)$, where $d = \dim(\mathfrak{u}(N)) = N^2$ and M is the number of closure iterations.

This algorithm computes the $\mathfrak{g}$-purity by first orthonormalizing the DLA basis under the trace inner product, then projecting the Hermitian operator H onto the complexified Lie algebra $\mathfrak{g}_{\mathbb{C}}$, and finally summing the squared magnitudes of the expansion coefficients.

Algorithm 2 Computable Algorithm for $\mathfrak{g}$-Purity

Require: Real linearly independent basis $\mathcal{B} = \{Y_1, \dots, Y_d\}$ of $\mathfrak{g} \subset \mathfrak{u}(N)$
Require: Hermitian operator $H \in \mathrm{Herm}(N)$
Ensure: $\mathfrak{g}$-purity $\mathcal{P}_{\mathfrak{g}}(H) \in \mathbb{R}_{\geq 0}$

1: $\mathcal{B}_{\text{ortho}} \leftarrow$ Schmidt orthonormalize $\mathcal{B}$ under $\langle A, B \rangle = \mathrm{Tr}(A^\dagger B)$
2: $\mathcal{P} \leftarrow 0$
3: **for** $B_j \in \mathcal{B}_{\text{ortho}}$ **do**
4: $\quad c_j \leftarrow \mathrm{Tr}\left(B_j^\dagger H\right)$
5: $\quad \mathcal{P} \leftarrow \mathcal{P} + |c_j|^2$
6: **end for**
7: **return** $\mathcal{P}$

The algorithm is numerically stable and runs in finite time. Orthogonalization uses Gram–Schmidt with trace inner product, which respects the complexification $\mathfrak{g}_{\mathbb{C}}$ automatically.

Time complexity: $O(d^2N^3 + dN^2)$, where $d = \dim(\mathfrak{g})$ and N is the Hilbert space dimension.

Remain Problem:

Given a compact real semisimple Lie algebra $\mathfrak{g} \subseteq \mathfrak{u}(N)$, decompose $\mathfrak{g}$ into a direct sum of **simple ideals** and compute the dimension of each simple summand $\dim \mathfrak{g}_j$.

3.2.2 Computing DLA of Classical Examples

To demonstrate the inductive closure algorithm in a nontrivial quantum control setting, we compute the dynamical Lie algebra for the transverse-field Ising model and so on.

Proposition 6 (Pauli Operator Algebra) *The single-qubit Pauli operators X_j, Y_j, Z_j satisfy the following identities:*

1. ***Multiplication Rules:***

$$X_jY_j = iZ_j, \quad Y_jZ_j = iX_j, \quad Z_jX_j = iY_j,$$
$$Y_jX_j = -iZ_j, \quad Z_jY_j = -iX_j, \quad X_jZ_j = -iY_j.$$

2. ***Commutation Relations:*** *$(j \neq k)$*

$$[X_j, Y_j] = 2iZ_j, \quad [Y_j, Z_j] = 2iX_j, \quad [Z_j, X_j] = 2iY_j,$$

$$\big[A_j, B_k\big] = 0, \quad \forall A, B \in \{X, Y, Z, I\}.$$

3. ***Skew-Hermitian generators:***

$$[iA_j, iB_j] = -[A_j, B_j], \quad j = k.$$

4. ***Splitting rules for nested Pauli strings ($j \neq k$):***

$$\begin{aligned} [A_j B_k,\, C_j] &= [A_j, C_j]\, B_k, \\ [A_j B_k,\, D_k] &= A_j\, [B_k, D_k] \\ [A_j B_k,\, A_j D_k] &= [B_k, D_k]\,. \end{aligned}$$

Example1: Nearest-Neighbor Ising Couplings

Let

$$\mathcal{G} = \big\{ Z_1 Z_2, Z_2 Z_3, \ldots, Z_{n-1} Z_n \big\}.$$

All Z-type Pauli operators commute mutually on distinct sites, so every pair of generators satisfies

$$[Z_j Z_{j+1},\, Z_k Z_{k+1}] = 0.$$

No new elements are generated by commutators.

$$\mathfrak{g}_{\mathrm{DLA}} = \mathrm{span}_{\mathbb{R}}\big\{\, i Z_j Z_{j+1} \,\big\}, \quad \dim \mathfrak{g}_{\mathrm{DLA}} = n - 1.$$

Example 2: Transverse-Field Ising Model on 2 Qubits

Let $n = 2$ (two qubits). The Hermitian generators are:

$$\mathcal{G} = \{X_1, X_2, Z_1 Z_2\}.$$

The dynamical Lie algebra:

$$\mathfrak{g}_{\mathrm{DLA}} = \langle iX_1, iX_2, iZ_1 Z_2 \rangle_{\mathbb{R},\mathrm{Lie}}.$$

1. Initialization:

$$\mathcal{B}_0 = \{iX_1, iX_2, iZ_1 Z_2\}, \quad \dim = 3.$$

2. First iteration:

$$\begin{aligned} [iX_1, iX_2] &= 0, \\ [iX_1, iZ_1 Z_2] &= 2iY_1 Z_2, \\ [iX_2, iZ_1 Z_2] &= 2iZ_1 Y_2. \end{aligned}$$

   New basis:

$$\mathcal{B}_1 = \{iX_1, iX_2, iZ_1 Z_2, iY_1 Z_2, iZ_1 Y_2\}, \quad \dim = 5.$$

3. Second iteration:

$$\begin{aligned} [iX_1, iY_1 Z_2] &= -2iZ_1 Z_2, \\ [iX_1, iZ_1 Y_2] &= 2iY_1 Y_2, \\ [iX_2, iY_1 Z_2] &\propto {\color{red} 2iY_1 Z_2}, \\ [iX_2, iZ_1 Y_2] &\propto iZ_1 Z_2, \\ [iZ_1 Z_2, iY_1 Z_2] &\propto iX_1, \\ [iZ_1 Z_2, iZ_1 Y_2] &\propto iX_2. \end{aligned}$$

   New basis:

$$\mathcal{B}_2 = \mathcal{B}_1 \cup \{iY_1 Y_2\}, \quad \dim = 6.$$

4. Closure:

$$\begin{aligned}
[iX_1, iY_1Y_2] &\propto iZ_1Y_2, \\
[iX_2, iY_1Y_2] &\propto iY_1Z_2, \\
[iZ_1Z_2, iY_1Y_2] &= 0, \\
[iY_1Z_2, iY_1Y_2] &\propto iX_2, \\
[iZ_1Y_2, iY_1Y_2] &\propto iX_1.
\end{aligned}$$

   All commutators remain in $\mathrm{span}_{\mathbb{R}}\,\mathcal{B}_2$. The algebra is closed.

5. Final result:

$$\boxed{\mathfrak{g}_{\mathrm{DLA}} = \mathrm{span}_{\mathbb{R}}\left\{iX_1,\, iX_2,\, iZ_1Z_2,\, iY_1Z_2,\, iZ_1Y_2,\, iY_1Y_2\right\}, \quad \dim = 6 = 3n}$$

3.2.3 Numerical computation

The implementation strictly follows Algorithm 1, with core logic visualized as:

1. Core modules & code mapping

Function	Numerical Task	Algorithm Step
`tensor`$(\cdot)$	Multi-qubit operator synthesis	Generator construction
`vec`$(\cdot)$	Complexreal vector embedding	Independence detection prep
`basis`$(\cdot)$	QR-based rank evaluation	Independent basis extraction
`closure_dim`$(\cdot)$	Commutator iteration + closure	Lie bracket closure loop

2. Computation flowchart

Algorithm 3 Numerical DLA Computation Flow

```
Input: Hermitian generator set 𝒢
1. Skew-Hermitian conversion: X ← iX for all X ∈ 𝒢
2. Tensor product expansion: Build multi-qubit operators
3. Embed operators to real vectors (real+imag parts)
4. Filter initial independent basis via QR factorization
5. Loop:
   a. Compute all pairwise commutators of current basis
   b. Merge commutators into operator set
   c. Refresh independent basis (QR factorization)
   d. Break if basis dimension unchanged
6. Output: Dimension of 𝔤_DLA
```

Complexity $\sim O(N^5) = O(2^{5n}) = O(32^n)$

3. Key numerical rules

- Only real linear independence is considered (consistent with $\mathbb{R}$-Lie algebra)
- QR factorization tolerance: 10^{-8} (ignores numerical noise)
- Termination condition: No new independent operators in consecutive iterations

Case 1: Transverse-Field Ising Model (Nearest-Neighbor)

$$\mathcal{G} = \{X_1, X_2, \ldots, X_n\} \cup \{Z_1Z_2, Z_2Z_3, \ldots, Z_{n-1}Z_n\}.$$

After running "Case1.py" , I got the following result:

Table 3.1: DLA dimensions of transverse-field Ising chain

Qubit number n	$\dim_{\mathbb{R}} \mathfrak{g}_{\text{DLA}}$
2	6
3	14
4	20
5	–

Case 2: X+ZZ form

$$\mathcal{G} = \{X_i \mid 1 \le i \le n\} \cup \{Z_i Z_j \mid 1 \le i < j \le n\}.$$

Table 3.2: DLA dimensions for $\boldsymbol{\mathcal{G} = \{X_i\} \cup \{Z_i Z_j \mid i < j\}}$

Qubit number n	$\dim_{\mathbb{R}} \mathfrak{g}_{\text{DLA}}$
2	6
3	18
4	40

Case 3: X+$Z_1..Z_n$ form

$$\mathcal{G} = \{X_i \mid 1 \le i \le n\} \cup \{Z_1 Z_2 ... Z_n\}.$$

Table 3.3: DLA dimensions for $\boldsymbol{\mathcal{G} = \{X_i\} \cup \{Z_1 Z_2 \cdots Z_n\}}$

Qubit number n	$\dim_{\mathbb{R}} \mathfrak{g}_{\text{DLA}}$
2	6
3	11
4	-

Problem: For general n, $\dim =?$

3.3 Pauli group and Optimizing for DLA computation

Definition 7 (Pauli Group) For n qubits system, we define the Pauli group by

$$\begin{aligned}\mathcal{P}_n &:= \left\{\alpha P | \alpha \in \{\pm 1, \pm i\}, P \in \{I, X, Y, Z\}^{\otimes n}\right\} \\ &= \left\{i^k X^{a_1} Z^{b_1} \otimes ... \otimes X^{a_n} Z^{b_n} | k \in \mathbb{Z}; a_j, b_j \in \{0, 1\}\right\}\end{aligned}$$

Where

$$\{I, X, Y, Z\}^{\otimes n} := \{P_1 \otimes P_2 \otimes ... \otimes P_n | P_j \in \{I, X, Y, Z\}\}$$

$$(X, Y, Z) = \left(\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\right)$$

Let's simplify the the multiplication rule :

$$X^aZ^bX^cZ^d = \begin{cases} X^{a+c}Z^d & b=0 \\ X^aZ^{b+d} & c=0 \\ X^aZXZ^d = X^a\left(-XZ\right)Z^d = -X^{a+1}Z^{d+1} & b=c=1 \end{cases}$$

We obtain the identity

$$X^aZ^bX^cZ^d = \left(-1\right)^{bc}X^{a+c}Z^{b+d}$$

Define a map $P : \mathbb{F}_2^{2n} \to \mathcal{P}_n$ by

$$P\left(v\right) = X^{a_1}Z^{b_1} \otimes ... \otimes X^{a_n}Z^{b_n} \quad \text{if} \quad v = \left(a_1, b_1, ..., a_n, b_n\right)$$

Then for $v_1 = \left(a_1, b_1, ..., a_n, b_n\right), v_2 = \left(c_1, d_1, ..., c_n, d_n\right)$, we have

$$\begin{aligned} P\left(v_1\right)P\left(v_2\right) &= \left(X^{a_1}Z^{b_1} \otimes ... \otimes X^{a_n}Z^{b_n}\right)\left(X^{c_1}Z^{d_1} \otimes ... \otimes X^{c_n}Z^{d_n}\right) \\ &= X^{a_1}Z^{b_1}X^{c_1}Z^{d_1} \otimes ... \otimes X^{a_n}Z^{b_n}X^{c_n}Z^{d_n} \\ &= \left(-1\right)^{b_1c_1+...+b_nc_n}X^{a_1+c_1}Z^{b_1+d_1} \otimes ... \otimes X^{a_n+c_n}Z^{b_n+d_n} \\ &= \left(-1\right)^{\omega(v_1,v_2)}X^{a_1+c_1}Z^{b_1+d_1} \otimes ... \otimes X^{a_n+c_n}Z^{b_n+d_n} \\ &= \left(-1\right)^{\omega(v_1,v_2)}P(v_1+v_2) \end{aligned}$$

where

$$\omega\left(v_1, v_2\right) = b_1c_1 + ... + b_nc_n = {v_1}^T\Omega v_2, \quad \Omega = \bigoplus_{j=1}^{n}\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

Then we have

$$\boxed{i^{k_1}P\left(v_1\right)i^{k_2}P\left(v_2\right) = i^{k_1+k_2+2\omega(v_1,v_2)}P\left(v_1+v_2\right)}$$

Lemma 8 *Let $V_1, \ldots, V_n, W_1, \ldots, W_n$ be k-modules, and for each $1 \leq j \leq n$, let $f_j : V_j \to W_j$ be a k-linear map. Then there exists a* ***unique*** *k-linear map*

$$F : V_1 \otimes V_2 \otimes \cdots \otimes V_n \longrightarrow W_1 \otimes W_2 \otimes \cdots \otimes W_n$$

satisfying

$$F(v_1 \otimes v_2 \otimes \cdots \otimes v_n) = f_1(v_1) \otimes f_2(v_2) \otimes \cdots \otimes f_n(v_n)$$

for all $v_j \in V_j$, $1 \leq j \leq n$. We denote this canonical map by

$$F = \bigotimes_{j=1}^{n} f_j = f_1 \otimes f_2 \otimes \cdots \otimes f_n.$$

Proof Using the following commutative diagram and the universal property.

$$\begin{tikzcd} V_1 \times ... \times V_n \arrow[r, "h_V"] \arrow[d, "f_1\times ...\times f_n"'] \arrow[dr, "f"] & V_1 \otimes ... \otimes V_n \arrow[d, dashed, "F=\tilde{f}"] \\ W_1 \times ... \times W_n \arrow[r, "h_W"] & W_1 \otimes ... \otimes W_n \end{tikzcd}$$

■

Definition 9 Let $\mathcal{H}_1, \dots, \mathcal{H}_n$ be finite-dimensional Hilbert spaces, and $\mathcal{B}(\mathcal{H}_j)$ the algebra of bounded operators on $\mathcal{H}_j$. Take the standard trace $\mathrm{tr}_j : \mathcal{B}(\mathcal{H}_j) \to \mathbb{C}$ on each component. For a subset $S \subseteq \{1, 2, \dots, n\}$, define a family of linear maps

$$\tau_j := \begin{cases} \mathrm{tr}_j, & j \in S, \\ \mathrm{id}_{\mathcal{B}(\mathcal{H}_j)}, & j \notin S. \end{cases}$$

By the previous lemma, there exists a unique linear map

$$\mathrm{tr}_S := \tau_1 \otimes \tau_2 \otimes \cdots \otimes \tau_n : \mathcal{B}(\mathcal{H}_1) \otimes ... \otimes \mathcal{B}(\mathcal{H}_n) \longrightarrow \bigotimes_{j \notin S} \mathcal{B}(\mathcal{H}_j)$$

called the **partial trace** over S, satisfying

$$\mathrm{tr}_S \left(\bigotimes_{j=1}^{n} A_j \right) = \left(\prod_{j \in S} \mathrm{tr}(A_j) \right) \bigotimes_{j \notin S} A_j$$

for all $A_j \in \mathcal{B}(\mathcal{H}_j)$.

Lemma 10 *The Pauli groups have the following properties.*

- *Let $\alpha, \beta \in \mathbb{C}^\times$ and $P, Q \in \{I, X, Y, Z\}^{\otimes n}$ such that $\alpha P = \beta Q$, then $P = Q$ and $\alpha = \beta$.*
- *The cardinality of the Pauli group is $|\mathcal{P}_n| = 4^{n+1}$.*
- *There is a bijection*
$$\mathbb{Z}_4 \times \mathbb{F}_2^{2n} = \left\{ (k, v) : k \in \mathbb{Z}_4,\ v \in \mathbb{F}_2^{2n} \right\} \xleftrightarrow{1:1} \mathcal{P}_n.$$

Proof Let
$$P = P_1 \otimes \cdots \otimes P_n, \quad Q = Q_1 \otimes \cdots \otimes Q_n, \quad Q_j, P_j \in \{I, X, Y, Z\}.$$
Since $X^2 = Y^2 = Z^2 = I^2 = I$, we have

$$Q^2 = Q_1^2 \otimes \cdots \otimes Q_n^2 = I \otimes \cdots \otimes I = I.$$

Thus $\beta^{-1} \alpha P Q = \beta^{-1} \beta Q^2 = I$. Note that

$$\beta^{-1} \alpha P Q = \beta^{-1} \alpha \bigotimes_{j=1}^{n} P_j Q_j = c \bigotimes_{j=1}^{n} R_j, \quad R_j \in \{I, X, Y, Z\},$$

where $P_j Q_j = I$ or $P_j Q_j = \pm i R_j$, so $c \neq 0$.

If there exists j_0 such that $R_{j_0} \neq I$, then $\mathrm{tr}(R_{j_0}) = 0$. Taking the partial trace over $\{j_0\}$ on both sides of $\beta^{-1} \alpha P Q = I$, we obtain

$$\mathrm{tr}_{\{j_0\}} (I) = \mathrm{tr}_{\{j_0\}} \left(\beta^{-1} \alpha P Q \right) = \mathrm{tr}_{\{j_0\}} \left(c \bigotimes_{j=1}^{n} R_j \right) = c \, \mathrm{tr} \left(R_{j_0} \right) \bigotimes_{j \neq j_0} R_j = 0.$$

This is a contradiction. Therefore, $P_j Q_j = I \implies P_j = Q_j$ for all j. So $P = Q \implies \alpha = \beta$.

As a corollary, we compute

$$\begin{aligned} |\mathcal{P}_n| &= \# \left\{ \alpha P \mid \alpha \in \{\pm 1, \pm i\},\ P \in \{I, X, Y, Z\}^{\otimes n} \right\} \\ &= \# \{\pm 1, \pm i\} \cdot \# \{I, X, Y, Z\}^{\otimes n} = 4^{n+1}. \end{aligned}$$

It is straightforward to check that the map

$$\Phi : (k, v) \mapsto i^k P(v)$$

is surjective from $\mathbb{Z}_4 \times \mathbb{F}_2^{2n}$ to $\mathcal{P}_n$. Since $|\mathcal{P}_n| = |\mathbb{Z}_4 \times \mathbb{F}_2^{2n}|$, the map is a bijection. ■

Theorem 11 *We define a group*

$$\mathbb{Z}_4 \ltimes_o \mathbb{F}_2^{2n} = \left\{(k, v) : k \in \mathbb{Z}_4, v \in \mathbb{F}_2^{2n}\right\}$$

by the multiplication

$$(k_1, v_1) * (k_2, v_2) := (k_1 + k_2 + 2\omega\left(v_1, v_2\right), v_1 + v_2)$$

Then the map $\Phi : (k, v) \mapsto i^k P\left(v\right)$*gives a group isomorphism*

$$\mathbb{Z}_4 \ltimes_o \mathbb{F}_2^{2n} \simeq \mathcal{P}_n.$$

Proof It can be proved by above lemma an the identies

$$i^{k_1} P\left(v_1\right) i^{k_2} P\left(v_2\right) = i^{k_1+k_2+2\omega(v_1,v_2)} P\left(v_1 + v_2\right)$$

which we calculated before. ■

Theorem 12 *The set*

$$\mathcal{B} = \left\{iP | P \in \{I, X, Y, Z\}^{\otimes n}\right\}$$

forms an $\mathbb{R}$*-base of* $\mathfrak{u}(2^n)$.

Proof Clearly every iP is skew-Hermitian, so

$$iP \in \mathfrak{u}(2^n).$$

Assume real coefficients λ_P satisfy

$$\sum_P \lambda_P \, iP = 0.$$

By the trace orthogonality of Pauli operators

$$\operatorname{tr}(PQ) = \begin{cases} 2^n, & P = Q, \\ 0, & P \neq Q, \end{cases}$$

multiply by Q and take trace:

$$0 = \left(\sum_{P\in\{I,X,Y,Z\}^{\otimes n}} \lambda_P \, iP\right) Q = \sum_{P\in\{I,X,Y,Z\}^{\otimes n}} \lambda_P i \, PQ$$

$$0 = \operatorname{tr}\left(\sum_{P\in\{I,X,Y,Z\}^{\otimes n}} \lambda_P i \, PQ\right) = \sum_{P\in\{I,X,Y,Z\}^{\otimes n}} \lambda_P i \operatorname{tr} PQ = \lambda_Q i 2^n = 0 \Longrightarrow \lambda_Q = 0, \quad \forall Q.$$

Thus $\{iP\}$ is $\mathbb{R}$-linearly independent.

Note

$$\left|\{I, X, Y, Z\}^{\otimes n}\right| = 4^n = \dim_{\mathbb{R}} \mathfrak{u}(2^n).$$

An independent set with full cardinality is a basis. ■

These two theorems establish a rigorous bridge between Pauli operators and algebraic structures. The first theorem encodes all Pauli multiplication and phase rules into a symplectic vector space over $\mathbb{F}_2^{2n}$. The second provides a canonical real basis for $\mathfrak{u}(2^n)$ formed by skew-Hermitian Pauli elements iP. Combined, they reduce operator commutation and linear independence to simple binary vector operations. This foundation drastically accelerates DLA closure computation and simplifies Lie algebraic analysis.

Since $Y = iXZ$, we know the base

$$\mathcal{B} = \left\{ iP | P \in \{I, X, iXZ, Z\}^{\otimes n} \right\} = \left\{ i^{1+\sigma(P)} P | P \in \{I, X, XZ, Z\}^{\otimes n} \right\} = \left\{ i^{1+\sigma(v)} P\left(v\right) | v \in \mathbb{F}_2^{2n} \right\}$$

where

$$\sigma\left(v\right) = \sum_j 1_{P_j = XZ} = \sum_j 1_{\{(a_j, b_j) = (1,1)\}} = \sum_j a_j b_j = \omega\left(v, v\right).$$

Denote $B(v) = i^{1+\omega(v,v)} P(v)$, then $\mathcal{B} = \left\{ B(v) | v \in \mathbb{F}_2^{2n} \right\}$, we have

$$\begin{aligned}
\left[B\left(v\right), B\left(w\right)\right] &= \left(i^{2+\sigma(v)+\sigma(w)+2\omega(v,w)} - i^{2+\sigma(v)+\sigma(w)+2\omega(w,v)} \right) P\left(v+w\right) \\
&= -i^{\sigma(v)+\sigma(w)} \left((-1)^{\omega(v,w)} - (-1)^{\omega(w,v)} \right) P\left(v+w\right) \\
&= \begin{cases} \pm 2 i^{\sigma(v)+\sigma(w)} P\left(v+w\right) & \omega\left(v,w\right) + \omega\left(w,v\right) \equiv 1 \bmod 2. \\ 0 & \omega\left(v,w\right) + \omega\left(w,v\right) \equiv 0 \bmod 2. \end{cases}
\end{aligned}$$

Since ω is bilinear,

$$\sigma\left(v+w\right) = \omega\left(v+w, v+w\right) = \sigma\left(v\right) + \sigma\left(w\right) + \omega\left(w,v\right) + \omega\left(v,w\right).$$

If $\omega\left(v,w\right) + \omega\left(w,v\right) \equiv 1 \bmod 2$, then

$$\sigma\left(v\right) + \sigma\left(w\right) \equiv \sigma\left(v+w\right) + \omega\left(w,v\right) + \omega\left(v,w\right) \equiv \sigma\left(v+w\right) + 1 \bmod 2$$

Thus

$$\left[B\left(v\right), B\left(w\right)\right] = \begin{cases} \pm 2 B\left(v+w\right) & \omega\left(v,w\right) + \omega\left(w,v\right) \equiv 1 \bmod 2. \\ 0 & \omega\left(v,w\right) + \omega\left(w,v\right) \equiv 0 \bmod 2. \end{cases}$$

Algorithm 4 Vector-Encoded Inductive Closure Algorithm for DLA

Require: Skew-Hermitian generator basis $\{B_1, B_2, \ldots, B_k\} \subset \mathfrak{u}(N)$
Ensure: Closed real basis $\mathcal{B}$ of the dynamical Lie algebra $\mathfrak{g}_{\mathrm{DLA}}$

```
Vector Encoding: Map each B_i to its unique index vector v_i ∈ F_2^{2n},
      and identify B_i ↔ B(v_i)
B ← distinct element subset of {B_1, B_2, ..., B_k}
while True do
  C ← {B(v + w) | B(v), B(w) ∈ B, ω(v, w) + ω(w, v) ≡ 1 (mod 2)}
  T ← B ∪ C
  B_new ← distinct element subset of T
  if |B_new| = |B| then
    B ← B_new
    break
  end if
  B ← B_new
end while
return B
```

3.4 Quantum Noise, Quantum Operations (Note of Chpat.8)

There are three classical definitions of quantum operation.

Definition 13 (System-Environment Model) Let ρ be a density matrix on system $\mathcal{H}_S$, and let $\mathcal{H}_E$ be an environment Hilbert space with initial state $|e_0\rangle\langle e_0|$. A **SEM quantum operation** $\mathcal{E}$ induced by system-environment interaction is defined as

$$\mathcal{E}(\rho) = \mathrm{tr}_E \left[U \left(\rho \otimes |e_0\rangle\langle e_0| \right) U^\dagger \right]$$

where U is a unitary operator on $\mathcal{H}_S \otimes \mathcal{H}_E$.

Definition 14 (Operator-Sum (Kraus) Representation) A map $\mathcal{E}$ is a **KRA quantum operation** if and only if there exists a set of linear operators $\{E_k\}$ (Kraus operators) such that

$$\mathcal{E}(\rho) = \sum_k E_k \rho E_k^\dagger, \qquad \sum_k E_k^\dagger E_k \leq I.$$

For trace-preserving operations (quantum channels), equality holds: $\sum_k E_k^\dagger E_k = I$.

Definition 15 (Axiomatic Definition) A linear map $\mathcal{E}$ is a **AXI quantum operation** if and only if it satisfies:

1. **Trace bound**: $0 \leq \mathrm{tr}(\mathcal{E}(\rho)) \leq \mathrm{tr}(\rho)$ for all states ρ;
2. **Convex linearity**: $\mathcal{E}\left(\sum p_i \rho_i\right) = \sum p_i \mathcal{E}(\rho_i)$;
3. **Complete positivity**: $\mathcal{I} \otimes \mathcal{E}$ maps positive operators to positive operators.

Theorem 16 *The three definitions of quantum operations (SEM, KRA, AXI) are equivalent.*

$$\text{SEM} \iff \text{KRA} \iff \text{AXI}$$

Proposition 17 *If $\mathcal{E}$ is an SEM quantum operation, then $\mathcal{E}$ is a KRA quantum operation.*

Proof Expand the unitary $U = \sum_i A_i \otimes |e_i\rangle\langle e_1|$. Then

$$U(X \otimes |e_1\rangle\langle e_1|)U^\dagger = \sum_{i,j} A_i X A_j^\dagger \otimes |e_i\rangle\langle e_j|.$$

Taking the partial trace over the environment gives

$$\mathcal{E}(X) = \sum_i A_i X A_i^*,$$

with $\sum_i A_i^\dagger A_i = I$ by unitarity. Thus $\mathcal{E}$ has a Kraus representation. ■

Proposition 18 *If $\mathcal{E}$ is a KRA quantum operation, then $\mathcal{E}$ is an AXI quantum operation.*

Proof Linearity holds by linearity of summation. For complete positivity, if $H \geq 0$ then

$$(\mathcal{E} \otimes I_k)(H) = \sum_i (K_i \otimes I) H (K_i \otimes I)^\dagger \geq 0.$$

For trace boundedness,

$$\mathrm{tr}(\mathcal{E}(X)) = \mathrm{tr}\left(X \sum_i K_i^\dagger K_i \right) \leq \mathrm{tr}(X).$$

Thus $\mathcal{E}$ satisfies all axioms. ■

Proposition 19 *If $\mathcal{E}$ is an AXI quantum operation, then $\mathcal{E}$ is an SEM quantum operation.*

Proof Let $J = (\mathcal{E} \otimes I_n)(|\Omega\rangle\langle\Omega|)$ be the Choi matrix. By complete positivity, $J \geq 0$. By spectral decomposition, $J = \sum_i \lambda_i v_i v_i^*$, which induces Kraus operators K_i. Construct an environment $\mathcal{H}_E$ and unitary U such that

$$\mathcal{E}(X) = \mathrm{tr}_E\left[U(X \otimes |e_0\rangle\langle e_0|)U^\dagger\right].$$

Thus $\mathcal{E}$ is an SEM quantum operation. ■

3.4.1 CPTP map (Quantum channe) and noisel

Definition 20 (CPTP Map / Quantum Channel) A linear map $\mathcal{E} : \mathcal{B}(\mathcal{H}) \to \mathcal{B}(\mathcal{H})$ is a *completely positive trace-preserving (CPTP) map*, if $\mathcal{E}$ is completely positive and $\mathrm{tr}(\mathcal{E}(\rho)) = \mathrm{tr}(\rho)$ for all density matrices ρ. All physical quantum noisy channels are CPTP maps.

$$\boxed{\textbf{Quantum Channel} = \textbf{CPTP map} \subsetneq \textbf{Quantum Operation}}$$

Definition 21 (Purity and noise) For a density matrix ρ, the *purity* is

$$\mathcal{P}(\rho) = \mathrm{tr}(\rho^2).$$

Pure state $\mathcal{P} = 1$, mixed state $\mathcal{P} < 1$.

Definition 22 (Quantum Noise) A CPTP map $\mathcal{E}$ is called *quantum noise* if $\mathcal{E}$ is non-unitary and **reduces the purity** of every non-maximally-mixed state:

$$\mathrm{tr}\big(\mathcal{E}(\rho)^2\big) < \mathrm{tr}\big(\rho^2\big), \quad \forall\, \rho \neq \tfrac{1}{d}I.$$

Quantum noise originates from systemenvironment entanglement and partial trace, leading to decoherence, information loss and growth of von Neumann entropy.

Definition 23 (von Neumann Entropy) The *von Neumann entropy* of ρ is

$$S(\rho) = -\,\mathrm{tr}\,\big(\rho \ln \rho\big).$$

It quantifies mixedness and disorder; quantum noise increases entropy.

Definition 24 (Trace Distance) For two density matrices ρ, σ,

$$D(\rho,\sigma) = \frac{1}{2}\|\rho - \sigma\|_1, \quad \|A\|_1 = \mathrm{tr}\Big(\sqrt{A^\dagger A}\Big).$$

Trace distance measures state distinguishability and noise-induced deviation.

Definition 25 (Fidelity)

$$F(\rho,\sigma) = \big\|\sqrt{\rho}\sqrt{\sigma}\big\|_1^2.$$

Fidelity quantifies state similarity; stronger noise leads to lower fidelity.

Definition 26 (Unitary Channel)

$$\mathcal{E}(\rho) = U\rho U^\dagger.$$

Describes ideal, **noise-free**, reversible unitary evolution with full coherence preserved.

Definition 27 (Bit Flip Channel)

$$\mathcal{E}(\rho) = (1-p)\rho + p\,X\rho X.$$

Models probabilistic computational basis flip $|0\rangle \leftrightarrow |1\rangle$, a typical classical-type quantum error.

Definition 28 (Phase Flip Channel)

$$\mathcal{E}(\rho) = (1-p)\rho + p\,Z\rho Z.$$

Induces random phase fluctuation, destroys off-diagonal coherence without changing diagonal populations.

Definition 29 (Depolarizing Channel)

$$\mathcal{E}(\rho) = (1-p)\rho + \frac{p}{3}\big(X\rho X + Y\rho Y + Z\rho Z\big).$$

Isotropic global noise that uniformly randomizes the qubit state and drives it toward the maximally mixed state.

Definition 30 (Pure Dephasing Channel)

$$\mathcal{E}\begin{pmatrix} \rho_{00} & \rho_{01} \\ \rho_{10} & \rho_{11} \end{pmatrix} = \begin{pmatrix} \rho_{00} & (1-\gamma)\rho_{01} \\ (1-\gamma)\rho_{10} & \rho_{11} \end{pmatrix}.$$

Pure decoherence channel suppressing off-diagonal elements, with energy populations unchanged.

Definition 31 (Amplitude Damping Channel)

$$\mathcal{E}(\rho) = K_0\rho K_0^\dagger + K_1\rho K_1^\dagger, \quad K_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix},\ K_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}.$$

Describes energy dissipation and spontaneous emission, modeling relaxation from excited state $|1\rangle$ to ground state $|0\rangle$.

Definition 32 (Fully Depolarizing Channel)

$$\mathcal{E}(\rho) = \frac{1}{2}I, \quad \forall\rho.$$

The extreme noise channel that completely erases quantum information and outputs the unique maximally mixed state.

3.5 Plan for Next Week

- Investigate the symbolic computation algorithm based on Pauli string generators.
- Derive general closed-form dimension formulas for Case 1 and Case 2.
- Study the decomposition algorithm of DLA into a direct sum of simple ideals and construct explicit bases for each component.
- Continue reading Chapter 8 and Chapter 9 of the textbook and finish corresponding exercises, alongside the study of Chapter 3 and Chapter 5.

3.6 Appendix

3.6.1 Solutions to exercises and problems

Theorem 33 (Polar Decomposition) *Let A be a linear operator on a finite-dimensional complex vector space V. Then there exist unitary operators U and positive operators J and K such that*

$$A = UJ = KU.$$

The positive operators J and K satisfying these equations are uniquely determined by

$$J \equiv \sqrt{A^\dagger A}, \qquad K \equiv \sqrt{AA^\dagger}.$$

Moreover, if A is invertible, then the unitary operator U is unique.

Proof We prove the left polar decomposition $A = UJ$. Define the positive operator

$$J = \sqrt{A^\dagger A}.$$

Since $A^\dagger A$ is positive semi-definite, its unique positive semi-definite square root J is well-defined.

For any $x \in V$, we have

$$\|Jx\|^2 = \langle Jx, Jx\rangle = \langle x, J^2 x\rangle = \langle x, A^\dagger A x\rangle = \|Ax\|^2.$$

This implies $\ker J = \ker A$, and the map

$$U : \operatorname{im} J \to \operatorname{im} A, \quad U(Jx) = Ax$$

is well-defined and isometric. Extend U unitarily to the whole space V by defining it on the orthogonal complement of $\operatorname{im} J$. Then U is unitary and $UJ = A$.

For uniqueness, suppose $A = U_1 J_1 = U_2 J_2$. Then

$$A^\dagger A = J_1^2 = J_2^2.$$

By the uniqueness of the positive square root, $J_1 = J_2 = \sqrt{A^\dagger A}$. If A is invertible, then J is invertible, so

$$U_1 = AJ^{-1} = U_2,$$

hence U is unique. The right decomposition $A = KU$ follows by applying the left decomposition to $A^\dagger$ and taking adjoints, with $K = \sqrt{AA^\dagger}$. ■

Theorem 34 (Singular Value Decomposition) *Let A be a linear operator on a finite-dimensional complex vector space V. Then there exist unitary operators U, V and a diagonal operator D with non-negative real entries such that*

$$A = UDV^\dagger.$$

The diagonal entries of D are the singular values of A, i.e., the non-negative square roots of the eigenvalues of $A^\dagger A$.

Proof By the polar decomposition, $A = UJ$ with U unitary and $J = \sqrt{A^\dagger A}$ positive. Since J is positive, it is diagonalizable in an orthonormal basis: there exists a unitary operator V such that $J = VDV^\dagger$, where D is diagonal with non-negative entries (the eigenvalues of J, i.e., the singular values of A). Substituting gives

$$A = UVDV^\dagger = (UV)DV^\dagger,$$

and UV is unitary. Renaming UV as U yields $A = UDV^\dagger$, as required. ■

Lemma 35 *Any single-qubit unitary operator U can be decomposed as*

$$U = e^{i\alpha} V = e^{i\alpha}\left(aI - i\vec{b}\cdot\vec{\sigma}\right),$$

with $\alpha \in \mathbb{R}$, $a \in \mathbb{R}$, $\vec{b} \in \mathbb{R}^3$, and $a^2 + \|\vec{b}\|^2 = 1$.

Proof Let U be a 2×2 unitary matrix. Since $|\det U| = 1$, we can write $\det U = e^{i2\alpha}$ for some real α. Define

$$V = e^{-i\alpha} U.$$

Then V is unitary and $\det V = 1$, so $V \in SU(2)$.

Any 2×2 matrix can be expressed as a linear combination of $I, \sigma_x, \sigma_y, \sigma_z$. For $V \in SU(2)$, unitarity and determinant 1 force the coefficients of I to be real and the coefficients of $\vec{\sigma}$ to be purely imaginary. Thus we can write

$$V = aI - i\vec{b}\cdot\vec{\sigma}$$

with $a \in \mathbb{R}$, $\vec{b} \in \mathbb{R}^3$, and $\det V = a^2 + \|\vec{b}\|^2 = 1$. Therefore

$$U = e^{i\alpha} V = e^{i\alpha}\left(aI - i\vec{b}\cdot\vec{\sigma}\right),$$

which completes the proof. ■

Theorem 36 *Suppose $\hat{m}$ and $\hat{n}$ are non-parallel real unit vectors in three dimensions, then any single-qubit unitary U can be written as*

$$U = e^{i\alpha} R_{\hat{n}}(\beta) R_{\hat{m}}(\gamma) R_{\hat{n}}(\delta),$$

for appropriate real numbers α, β, γ and δ.

Proof Let $c_\theta = \cos(\theta/2)$ and $s_\theta = \sin(\theta/2)$. By definition,

$$R_{\hat{u}}(\theta) = c_\theta I - i s_\theta \hat{u}\cdot\vec{\sigma}.$$

We first expand the product

$$R_{\hat{n}}(\beta) R_{\hat{m}}(\gamma) R_{\hat{n}}(\delta).$$

Compute the product of the first two factors:

$$\begin{aligned} R_{\hat{n}}(\beta) R_{\hat{m}}(\gamma) &= (c_\beta I - i s_\beta \hat{n}\cdot\vec{\sigma})(c_\gamma I - i s_\gamma \hat{m}\cdot\vec{\sigma}) \\ &= c_\beta c_\gamma I - i c_\beta s_\gamma \hat{m}\cdot\vec{\sigma} - i c_\gamma s_\beta \hat{n}\cdot\vec{\sigma} - s_\beta s_\gamma (\hat{n}\cdot\vec{\sigma})(\hat{m}\cdot\vec{\sigma}). \end{aligned}$$

Using the identity $(\hat{n}\cdot\vec{\sigma})(\hat{m}\cdot\vec{\sigma}) = (\hat{n}\cdot\hat{m})I + i(\hat{n}\times\hat{m})\cdot\vec{\sigma}$, we obtain

$$\begin{aligned} R_{\hat{n}}(\beta) R_{\hat{m}}(\gamma) = &\big(c_\beta c_\gamma - s_\beta s_\gamma \hat{n}\cdot\hat{m}\big) I \\ &- i\big(c_\beta s_\gamma \hat{m} + c_\gamma s_\beta \hat{n} - s_\beta s_\gamma \hat{n}\times\hat{m}\big)\cdot\vec{\sigma}. \end{aligned}$$

Now multiply by $R_{\hat{n}}(\delta) = c_\delta I - i s_\delta \hat{n}\cdot\vec{\sigma}$:

$$\begin{aligned} &R_{\hat{n}}(\beta) R_{\hat{m}}(\gamma) R_{\hat{n}}(\delta) \\ &= \big(c_\beta c_\gamma - s_\beta s_\gamma \hat{n}\cdot\hat{m}\big) c_\delta I - i\big(c_\beta c_\gamma - s_\beta s_\gamma \hat{n}\cdot\hat{m}\big) s_\delta \hat{n}\cdot\vec{\sigma} \\ &\quad - i c_\delta\big(c_\beta s_\gamma \hat{m} + c_\gamma s_\beta \hat{n} - s_\beta s_\gamma \hat{n}\times\hat{m}\big)\cdot\vec{\sigma} \\ &\quad - s_\delta\big(c_\beta s_\gamma \hat{m} + c_\gamma s_\beta \hat{n} - s_\beta s_\gamma \hat{n}\times\hat{m}\big)\cdot\vec{\sigma}\ \hat{n}\cdot\vec{\sigma}. \end{aligned}$$

Applying the Pauli identity again to the last term yields

$$\begin{aligned}&\big(c_\beta s_\gamma \hat{m} + c_\gamma s_\beta \hat{n} - s_\beta s_\gamma \hat{n}\times\hat{m}\big)\cdot\vec{\sigma}\;\hat{n}\cdot\vec{\sigma}\\ &= \big(c_\beta s_\gamma \hat{m}\cdot\hat{n} + c_\gamma s_\beta\big)I + i\big(c_\beta s_\gamma \hat{m}\times\hat{n} - s_\beta s_\gamma(\hat{n}\times\hat{m})\times\hat{n}\big)\cdot\vec{\sigma}.\end{aligned}$$

Substituting back and collecting terms, the entire product takes the form

$$AI - i\vec{B}\cdot\vec{\sigma},$$

where $A\in\mathbb{R}$ and $\vec{B}\in\mathbb{R}^3$ satisfy $A^2+\|\vec{B}\|^2=1$.

Since $\hat{m}$ and $\hat{n}$ are non-parallel, $\hat{n},\hat{m},\hat{n}\times\hat{m}$ form a basis of $\mathbb{R}^3$. The three angles β,γ,δ provide three real degrees of freedom that can be chosen to match any desired $\vec{B}$. The scalar A is then determined by the normalization condition.

Any single-qubit unitary U can be written as $U=e^{i\alpha}(AI-i\vec{B}\cdot\vec{\sigma})$. Thus there exist real numbers $\alpha,\beta,\gamma,\delta$ such that

$$U=e^{i\alpha}R_{\hat{n}}(\beta)R_{\hat{m}}(\gamma)R_{\hat{n}}(\delta),$$

which completes the proof. ■

Problem 1 (Functions of the Pauli matrices) Let $f\in H(\mathbb{C})$ be an analytic function, and $\vec{n}$ be a normalized vector in three dimensions, and let θ be real. Show that

$$f(\theta\vec{n}\cdot\vec{\sigma}) = \frac{f(\theta)+f(-\theta)}{2}I + \frac{f(\theta)-f(-\theta)}{2}\vec{n}\cdot\vec{\sigma}.$$

Proof Let $U=\vec{n}\cdot\vec{\sigma}=n_x\sigma_x+n_y\sigma_y+n_z\sigma_z$. We first verify that $U^2=I$. Using the Pauli multiplication identity

$$(\vec{n}\cdot\vec{\sigma})(\vec{n}\cdot\vec{\sigma}) = (\vec{n}\cdot\vec{n})I + i(\vec{n}\times\vec{n})\cdot\vec{\sigma},$$

and since $\vec{n}$ is a unit vector we have $\vec{n}\cdot\vec{n}=1$, while $\vec{n}\times\vec{n}=0$. Thus

$$U^2=(\vec{n}\cdot\vec{\sigma})^2=I.$$

It follows that for all integers $k\geq 0$,

$$U^{2k}=I,\qquad U^{2k+1}=U.$$

Since f is analytic, we may write its power series

$$f(z)=\sum_{k=0}^{\infty}a_k z^k.$$

Substituting $z=\theta U$ gives

$$f(\theta U)=\sum_{k=0}^{\infty}a_k\theta^k U^k = \sum_{m=0}^{\infty}a_{2m}\theta^{2m}U^{2m} + \sum_{m=0}^{\infty}a_{2m+1}\theta^{2m+1}U^{2m+1}.$$

Using $U^{2m}=I$ and $U^{2m+1}=U$,

$$f(\theta U)=\left(\sum_{m=0}^{\infty}a_{2m}\theta^{2m}\right)I + \left(\sum_{m=0}^{\infty}a_{2m+1}\theta^{2m+1}\right)U.$$

The even and odd parts of f satisfy

$$\frac{f(\theta)+f(-\theta)}{2}=\sum_{m=0}^{\infty}a_{2m}\theta^{2m},\qquad \frac{f(\theta)-f(-\theta)}{2}=\sum_{m=0}^{\infty}a_{2m+1}\theta^{2m+1}.$$

Thus

$$f(\theta\vec{n}\cdot\vec{\sigma}) = \frac{f(\theta)+f(-\theta)}{2}I + \frac{f(\theta)-f(-\theta)}{2}\vec{n}\cdot\vec{\sigma},$$

as required. ■

Theorem 37 (Schmidt Decomposition) *Suppose $|\psi\rangle$ is a pure state of a composite system AB, then there exist orthonormal states $\{|i\rangle_A\}_{i=1}^{d_A}$ for system A, orthonormal states $\{|i\rangle_B\}_{i=1}^{d_B}$ for system B, and real numbers $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_r > 0$ such that*

$$|\psi\rangle = \sum_{i=1}^{r} \lambda_i |i\rangle_A \otimes |i\rangle_B, \quad \sum_{i=1}^{r} \lambda_i^2 = 1.$$

The numbers λ_i are called the ***Schmidt coefficients*** *of $|\psi\rangle$, the number r is called the* ***Schmidt number*** *of $|\psi\rangle$, denoted by $r = \mathrm{Sch}(\psi)$.*

Proof Let $\{|j\rangle_A\}_{j=1}^{d_A}$ be an orthonormal basis for $\mathcal{H}_A$, and $\{|k\rangle_B\}_{k=1}^{d_B}$ be an orthonormal basis for $\mathcal{H}_B$. Then $|\psi\rangle$ can be uniquely expressed as

$$|\psi\rangle = \sum_{j=1}^{d_A} \sum_{k=1}^{d_B} a_{jk} |j\rangle_A \otimes |k\rangle_B,$$

where $a = (a_{jk})$ is a $d_A \times d_B$ complex matrix, and $\langle\psi|\psi\rangle = 1$ implies $\sum_{j,k} |a_{jk}|^2 = 1$.

By the singular value decomposition (SVD) for rectangular matrices, there exist a $d_A \times d_A$ unitary matrix U, a $d_B \times d_B$ unitary matrix V, and a $d_A \times d_B$ diagonal matrix $D = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_r, 0, \ldots, 0)$ with non-negative real entries $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_r > 0$ ($r = \mathrm{rank}(a) \leq \min(d_A, d_B)$), such that

$$a = UDV^\dagger.$$

Substitute the SVD into the expansion of $|\psi\rangle$:

$$\begin{aligned}
|\psi\rangle &= \sum_{j,k} (UDV^\dagger)_{jk} |j\rangle_A \otimes |k\rangle_B \\
&= \sum_{j,k} \sum_{i=1}^{r} U_{ji} \lambda_i (V^\dagger)_{ik} |j\rangle_A \otimes |k\rangle_B \\
&= \sum_{i=1}^{r} \lambda_i \left(\sum_{j=1}^{d_A} U_{ji} |j\rangle_A \right) \otimes \left(\sum_{k=1}^{d_B} V_{ik}^* |k\rangle_B \right).
\end{aligned}$$

Define the states for systems A and B:

$$|i\rangle_A \equiv \sum_{j=1}^{d_A} U_{ji} |j\rangle_A, \quad |i\rangle_B \equiv \sum_{k=1}^{d_B} V_{ik}^* |k\rangle_B, \quad \forall i = 1, 2, \ldots, r.$$

We extend $\{|i\rangle_A\}$ to a full orthonormal basis of $\mathcal{H}_A$ (for $i > r$) and $\{|i\rangle_B\}$ to a full orthonormal basis of $\mathcal{H}_B$ (for $i > r$), which is always possible by the Gram-Schmidt process.

By the unitarity of U and V, we verify the orthonormality:

$$\langle i|_A |j\rangle_A = \sum_{l,m} U_{li}^* U_{mj} \langle l|m\rangle_A = \sum_l U_{li}^* U_{lj} = (U^\dagger U)_{ij} = \delta_{ij},$$

$$\langle i|_B |j\rangle_B = \sum_{l,m} V_{il} V_{jm}^* \langle l|m\rangle_B = \sum_l V_{il} V_{jl}^* = (VV^\dagger)_{ij} = \delta_{ij}.$$

Substituting back, we obtain the Schmidt decomposition:

$$|\psi\rangle = \sum_{i=1}^{r} \lambda_i |i\rangle_A \otimes |i\rangle_B.$$

The normalization condition $\sum_{i=1}^{r} \lambda_i^2 = 1$ follows directly from $\langle\psi|\psi\rangle = 1$ and the orthonormality. ■

Corollary 38 *The Schmidt number r of $|\psi\rangle$ is equal to the rank of the coefficient matrix $a = (a_{jk})$, i.e.,*

$$\mathrm{Sch}(\varphi) = \mathrm{rank}(a^{\varphi}).$$

Theorem 39 (Properties of the Schmidt Number) *Let $|\psi\rangle$ be a pure state of a bipartite system AB, with Schmidt number $\mathrm{Sch}(\psi)$.*

1. *The Schmidt number equals the rank of the reduced density matrix:*

$$\mathrm{Sch}(\psi) = \mathrm{rank}(\rho_A) = \mathrm{rank}(\rho_B),$$

   where $\rho_A = \mathrm{tr}_B(|\psi\rangle\langle\psi|)$ and $\rho_B = \mathrm{tr}_A(|\psi\rangle\langle\psi|)$.
2. *If $|\psi\rangle = \sum_{j=1}^{N} |\alpha_j\rangle \otimes |\beta_j\rangle$ for some states $|\alpha_j\rangle$ of A and $|\beta_j\rangle$ of B, then $N \geq \mathrm{Sch}(\psi)$.*
3. *For any superposition $|\psi\rangle = \alpha|\varphi\rangle + \beta|\gamma\rangle$,*

$$\mathrm{Sch}(\psi) \geq \big|\mathrm{Sch}(\varphi) - \mathrm{Sch}(\gamma)\big|.$$

Proof

1. Let r be the Schmidt number of $|\psi\rangle$. By the Schmidt decomposition,

$$|\psi\rangle = \sum_{i=1}^{r} \lambda_i |i\rangle_A \otimes |i\rangle_B, \quad \lambda_i > 0, \sum_i \lambda_i^2 = 1$$

   with $\{|i\rangle_A\}, \{|i\rangle_B\}$ orthonormal bases. The reduced density matrix is

$$\rho_A = \mathrm{tr}_B\left(|\psi\rangle\langle\psi|\right) = \sum_{i,j=1}^{r} \lambda_i\lambda_j |i\rangle_A\langle j|_A \,\mathrm{tr}_B(|i\rangle_B\langle j|_B) = \sum_{i=1}^{r} \lambda_i^2 |i\rangle_A\langle i|_A,$$

   Since $\mathrm{tr}_B(|i\rangle_B\langle j|_B) = \delta_{ij}$. Because $\lambda_i > 0$, we know $\mathrm{rank}(\rho_A) = r = \mathrm{Sch}(\psi)$.
2. Suppose $|\psi\rangle = \sum_{j=1}^{N} |\alpha_j\rangle|\beta_j\rangle$ with un-normalized states $|\alpha_j\rangle \in \mathcal{H}_A$ and $|\beta_j\rangle \in \mathcal{H}_B$. First compute the reduced density matrix:

$$|\psi\rangle\langle\psi| = \sum_{j,k=1}^{N} |\alpha_j\rangle\langle\alpha_k| \otimes |\beta_j\rangle\langle\beta_k|.$$

   Taking partial trace over B, we get

$$\rho_A = \mathrm{tr}_B\left(|\psi\rangle\langle\psi|\right) = \sum_{j,k=1}^{N} \langle\beta_k|\beta_j\rangle\, |\alpha_j\rangle\langle\alpha_k|.$$

   so $\mathrm{rank}(\rho_A) \leq N$. From part (1), $\mathrm{rank}(\rho_A) = \mathrm{Sch}(\psi)$, hence $N \geq \mathrm{Sch}(\psi)$.
3. We know $\mathrm{rank}(X+Y) \geq |\,\mathrm{rank}(X) - \mathrm{rank}(Y)|$ for matrices. By the corollary, for any bipartite pure state $|\phi\rangle$, let a^{ϕ} be its coefficient matrix in a product basis, then $\mathrm{Sch}(\phi) = \mathrm{rank}(a^{\phi})$. For $|\psi\rangle = \alpha|\varphi\rangle + \beta|\gamma\rangle$, its coefficient matrix $a^{\psi} = \alpha a^{\varphi} + \beta a^{\gamma}$. Thus:

$$\mathrm{rank}(a^{\psi}) = \mathrm{rank}(\alpha a^{\varphi} + \beta a^{\gamma}) \geq |\,\mathrm{rank}(\alpha a^{\varphi}) - \mathrm{rank}(\beta a^{\gamma})| = |\,\mathrm{rank}(a^{\varphi}) - \mathrm{rank}(a^{\gamma})|.$$

   By $\mathrm{Sch}(\psi) = \mathrm{rank}(a^{\psi})$, $\mathrm{Sch}(\varphi) = \mathrm{rank}(a^{\varphi})$, $\mathrm{Sch}(\gamma) = \mathrm{rank}(a^{\gamma})$, we get:

$$\mathrm{Sch}(\psi) \geq |\,\mathrm{Sch}(\varphi) - \mathrm{Sch}(\gamma)|.$$

■

Problem 2 (Tsirelson's inequality) Suppose $Q = \vec{q} \cdot \vec{\sigma}$, $R = \vec{r} \cdot \vec{\sigma}$, $S = \vec{s} \cdot \vec{\sigma}$, $T = \vec{t} \cdot \vec{\sigma}$, where $\vec{q}, \vec{r}, \vec{s}, \vec{t}$ are real unit vectors in $\mathbb{R}^3$. Show that

$$(Q \otimes S + R \otimes S + R \otimes T - Q \otimes T)^2 = 4I + [Q, R] \otimes [S, T].$$

Use this result to prove that

$$\langle Q \otimes S \rangle + \langle R \otimes S \rangle + \langle R \otimes T \rangle - \langle Q \otimes T \rangle \leq 2\sqrt{2}.$$

Proof Let

$$C = Q \otimes S + R \otimes S + R \otimes T - Q \otimes T.$$

Step 1: Prove the operator identity.

Rewrite

$$C = (Q + R) \otimes S + (R - Q) \otimes T.$$

Expanding the square:

$$C^2 = (Q + R)^2 \otimes S^2 + (Q + R)(R - Q) \otimes ST + (R - Q)(Q + R) \otimes TS + (R - Q)^2 \otimes T^2.$$

Since $Q^2 = R^2 = S^2 = T^2 = I$, we have

$$(Q + R)^2 = 2I + QR + RQ, \qquad (R - Q)^2 = 2I - (QR + RQ).$$

Also

$$(Q + R)(R - Q) = [Q, R], \qquad (R - Q)(Q + R) = -[Q, R].$$

Thus

$$C^2 = \big(2I + QR + RQ\big) \otimes I + [Q, R] \otimes ST - [Q, R] \otimes TS + \big(2I - QR - RQ\big) \otimes I.$$

The anticommutator terms cancel, leaving

$$C^2 = 4I + [Q, R] \otimes (ST - TS) = 4I + [Q, R] \otimes [S, T].$$

Step 2: Prove the inequality.

Lemma 40 *Let A, B be Hermitian operators and C Hermitian.*

1. *For any normalized state $|\psi\rangle$, $\langle C \rangle \leq \|C\|_{\text{op}}$.*
2. $\|A \otimes B\|_{\text{op}} = \|A\|_{\text{op}} \|B\|_{\text{op}}$.
3. $\|[A, B]\|_{\text{op}} \leq 2\|A\|_{\text{op}} \|B\|_{\text{op}}$.

Since C is Hermitian,

$$\|C\|_{\text{op}}^2 = \|C^2\|_{\text{op}}.$$

By triangle inequality and Lemma (2):

$$\|C^2\|_{\text{op}} \leq \|4I\|_{\text{op}} + \big\|[Q, R] \otimes [S, T]\big\|_{\text{op}} = 4 + \|[Q, R]\|_{\text{op}} \|[S, T]\|_{\text{op}}.$$

By Lemma (3), $\|[Q, R]\|_{\text{op}} \leq 2$ and $\|[S, T]\|_{\text{op}} \leq 2$, so

$$\|C^2\|_{\text{op}} \leq 4 + 2 \cdot 2 = 8.$$

Hence

$$\|C\|_{\mathrm{op}} \le 2\sqrt{2}.$$

By Lemma

$$\langle C\rangle \le \|C\|_{\mathrm{op}} \le 2\sqrt{2},$$

which completes the proof. ■

Problem 3 (Circuit Conjugation and Commutation Identities) Let C denote the two-qubit CNOT gate with qubit 1 as the control and qubit 2 as the target. For any single-qubit operator A, write $A_1 = A \otimes I$ and $A_2 = I \otimes A$. Prove

$$CX_1C = X_1X_2, \quad CY_1C = Y_1X_2, \quad CZ_1C = Z_1,$$
$$CX_2C = X_2, \quad CY_2C = Z_1Y_2, \quad CZ_2C = Z_1Z_2,$$
$$R_{z,1}(\theta)C = CR_{z,1}(\theta), \quad R_{x,2}(\theta)C = CR_{x,2}(\theta).$$

Proof We first establish necessary structural lemmas for CNOT and operator conjugation.

Lemma 41 *The CNOT gate C is involutory, i.e., $C^2 = I$, and acts on the computational basis as*

$$C|a,b\rangle = \big|a, b\oplus a\big\rangle, \qquad \forall a,b \in \{0,1\}.$$

In particular, $C^{-1} = C$ and $C^\dagger = C$.

Lemma 42 *Let X, Y, Z be Pauli matrices. Conjugation by CNOT satisfies*

1. $C\big(P\otimes I\big)C = \begin{cases} P\otimes X, & P\in\{X,Y\},\\ P\otimes I, & P = Z;\end{cases}$
2. $C\big(I\otimes Q\big)C = \begin{cases} I\otimes X, & Q = X,\\ Z\otimes Q, & Q\in\{Y,Z\}.\end{cases}$

Lemma 43 *If bounded operators satisfy $AB = BA$, then for any analytic function f,*

$$f(A)B = Bf(A).$$

Matrix exponentials of commuting operators mutually commute.

We verify the Pauli conjugation in lemma42 on the complete computational basis. For control-qubit operators, direct basis computation gives

$$C(X\otimes I)C|a,b\rangle = |1-a, 1-b\rangle = (X\otimes X)|a,b\rangle,$$

$$C(Y\otimes I)C|a,b\rangle = (Y\otimes X)|a,b\rangle, \quad C(Z\otimes I)C|a,b\rangle = (Z\otimes I)|a,b\rangle.$$

For target-qubit operators,

$$C(I\otimes X)C|a,b\rangle = |a, 1-b\rangle = (I\otimes X)|a,b\rangle,$$

$$C(I\otimes Y)C|a,b\rangle = (Z\otimes Y)|a,b\rangle, \quad C(I\otimes Z)C|a,b\rangle = (Z\otimes Z)|a,b\rangle.$$

Linear operators coinciding on a full basis are identical, so all relations in lemma42 hold.

The six Pauli conjugation identities follow directly from lemma42. Substituting P, Q recovers

$$CX_1C = X_1X_2, \quad CY_1C = Y_1X_2, \quad CZ_1C = Z_1,$$

$$CX_2C = X_2, \quad CY_2C = Z_1Y_2, \quad CZ_2C = Z_1Z_2.$$

By lemma41, $C^2 = I$, so $CUC = U$ implies $UC = CU$. From $CZ_1C = Z_1$, we obtain $Z_1C = CZ_1$. The rotation $R_{z,1}(\theta) = e^{-i\theta Z_1/2}$ is an analytic function of Z_1. Applying lemma43, $R_{z,1}(\theta)$ commutes with C, hence

$$R_{z,1}(\theta)C = CR_{z,1}(\theta).$$

Similarly, $CX_2C = X_2$ yields $X_2C = CX_2$. Since $R_{x,2}(\theta) = e^{-i\theta X_2/2}$ is a function of X_2, lemma43 gives

$$R_{x,2}(\theta)C = CR_{x,2}(\theta).$$

All conjugation and commutation relations are thus proven. ■

Let U, V be bounded operators on a finite-dimensional Hilbert space. Then we have the folowing lemma:

Lemma 44

$$e^U e^V - e^{U+V} = \int_0^1 e^{Us}\left[U, e^{Vs}\right] e^{(U+V)(1-s)}\mathrm{d}s$$

Proof Let $F(s) = e^{Us}e^{Vs}e^{-(U+V)s}$, then

$$\begin{aligned} F'(s) &= Ue^{Us}e^{Vs}e^{-(U+V)s} + e^{Us}Ve^{Vs}e^{-(U+V)s} - e^{Us}e^{Vs}(U+V)e^{-(U+V)s} \\ &= e^{Us}\left[U, e^{Vs}\right] e^{-(U+V)s} \end{aligned}$$

So

$$F(1) - F(0) = e^U e^V e^{-(U+V)} - I = \int_0^1 e^{Us}\left[U, e^{Vs}\right] e^{-(U+V)s}\mathrm{d}s.$$

■

Lemma 45

$$[U, e^{sV}] = e^{sV}\int_0^s e^{-Vt}G(t)\mathrm{d}t, \quad G(s) = [U, V]e^{Vs}$$

Proof For $g(s) = [U, e^{Vs}] = Ue^{Vs} - e^{Vs}U$ we have

$$\begin{aligned} \frac{\mathrm{d}g}{\mathrm{d}s} &= UVe^{Vs} - Ve^{Vs}U \\ &= UVe^{Vs} - VUe^{Vs} + VUe^{Vs} - Ve^{Vs}U \\ &= [U, V]\,e^{Vs} + V\left[U, e^{Vs}\right] = G(s) + Vg(s) \end{aligned}$$

For $f(s) = e^{Vs}\int_0^s e^{-Vt}G(t)\mathrm{d}t$, we have

$$\begin{aligned} \frac{\mathrm{d}f}{\mathrm{d}s} &= \frac{\mathrm{d}e^{Vs}}{\mathrm{d}s}\int_0^s e^{-Vt}G(t)\mathrm{d}t \,+\, e^{Vs}\frac{\mathrm{d}}{\mathrm{d}s}\int_0^s e^{-Vt}G(t)\mathrm{d}t \\ &= Ve^{Vs}\int_0^s e^{-Vt}G(t)\mathrm{d}t \,+\, e^{Vs}e^{-Vs}G(s) \\ &= V\left(e^{Vs}\int_0^s e^{-Vt}G(t)\mathrm{d}t\right) + G(s) = Vf(s) + G(s) \end{aligned}$$

Upon direct computation, $f(0) = g(0) = 0$. By the uniqueness of solutions to linear ordinary differential equations, we conclude $f(s) = g(s)$. ■

Lemma 46

$$\|[U, e^{sV}]\| \leq s\,\|[U,V]\|\,\|e^{sV}\|$$

Proof

$$\begin{aligned}
\|f(s)\| &= \left\| \int_0^s e^{V(s-t)}\,[U,V]\,e^{Vt}\mathrm{d}t \right\| \\
&\leq \int_0^s \|\, e^{V(s-t)} \,\|\ \|\,[U,V]\,\|\,\|\,e^{Vt}\,\|\ \mathrm{d}t \\
&\leq \left\|e^{Vs}\right\| \int_0^s \|\,[U,V]\,\|\,\mathrm{d}t = s\|[U,V]\|\,\left\|e^{Vs}\right\|
\end{aligned}$$

■

Lemma 47

$$\left\|e^U e^V - e^{U+V}\right\| \leq \tfrac{1}{2}\,\|[U,V]\|\,e^{\|U\|+\|V\|}.$$

Proof

$$\begin{aligned}
\left\|e^U e^V - e^{U+V}\right\| &= \left\| \int_0^1 e^{Us}\,\left[U, e^{Vs}\right] e^{(U+V)(1-s)}\mathrm{d}s \right\| \\
&\leq \int_0^1 \left\|e^{Us}\right\| \cdot \left\|\left[U, e^{Vs}\right]\right\| \cdot \left\|e^{(U+V)(1-s)}\right\| \mathrm{d}s \\
&\leq \int_0^1 e^{\|U\|+(1-s)\|V\|} s\ \|\,[U,V]\,\|\ \left\|e^{Vs}\right\| \mathrm{d}s \\
&\leq e^{\|U\|}\,\|[U,V]\|\,e^{\|V\|} \int_0^1 s\mathrm{d}s \\
&= \frac{1}{2}\,\|[U,V]\|\,e^{\|U\|+\|V\|}
\end{aligned}$$

■

Lemma 48 *Let $X, Y \in \mathcal{B}(\mathcal{H})$ be bounded linear operators. Then*

$$\|X^n - Y^n\| \leq n\ \max\left\{\|X\|, \|Y\|\right\}^{n-1}\|X - Y\|.$$

In particular, if $\|X\| = \|Y\| = 1$, we have

$$\|X^n - Y^n\| \leq n\,\|X - Y\|.$$

Proof We use the telescoping decomposition

$$X^n - Y^n = \sum_{k=0}^{n-1} X^k (X - Y) Y^{n-1-k}.$$

By triangle inequality and submultiplicativity of operator norm,

$$\begin{aligned}
\|X^n - Y^n\| &\leq \sum_{k=0}^{n-1} \|X^k\|\,\|X-Y\|\,\|Y^{n-1-k}\| \\
&\leq \|X - Y\| \sum_{k=0}^{n-1} \|X\|^k \|Y\|^{n-1-k} \\
&\leq n\ \max\left\{\|X\|, \|Y\|\right\}^{n-1}\|X - Y\|.
\end{aligned}$$

When $\|X\| = \|Y\| = 1$, the higher-order terms vanish, which yields $\|X^n - Y^n\| \leq n\|X - Y\|$. ■

Theorem 49 *Let A, B be Hermitian matrices, we have Trotter formula with explicit uniform remainder upper bound:*

$$\left\| \left(e^{itA/n} e^{itB/n}\right)^n - e^{it(A+B)} \right\| \le \frac{t^2}{2n} \, \|[A, B]\| \, e^{\frac{|t|}{n}(\|A\|+\|B\|)}.$$

Proof Set $U = \frac{it}{n}A, \quad V = \frac{it}{n}B$ in Lemma 47, for each single step

$$\left\|e^{\frac{it}{n}A} e^{\frac{it}{n}B} - e^{\frac{it}{n}(A+B)}\right\| \le \frac{t^2}{2n^2} \|[A, B]\| e^{\frac{|t|}{n}(\|A\|+\|B\|)}.$$

Write

$$E_n = e^{\frac{it}{n}A} e^{\frac{it}{n}B}, \quad S_n = e^{\frac{it}{n}(A+B)}.$$

By the telescoping identity for operator powers

$$E_n^n - S_n^n = \sum_{k=0}^{n-1} E_n^k (E_n - S_n) S_n^{n-1-k},$$

and the unitarity of E_n, S_n yields $\|E_n^k\| = \|S_n^k\| = 1$. Therefore

$$\|E_n^n - S_n^n\| \le n \, \|E_n - S_n\|.$$

Substitute the local bound:

$$\|E_n^n - S_n^n\| \le n \cdot \frac{t^2}{2n^2} \|[A, B]\| e^{\frac{|t|}{n}(\|A\|+\|B\|)}.$$

we obtain the uniform upper bound. ■

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